

Definition *directly_committed* (*net* : *network*) (*e* : *entry*) :=

$\exists$ *quorum*,
 $NoDup$ *quorum* $\wedge$
 $length$ *quorum* $>$ $div2$ ($length$ *nodes*) $\wedge$
 $\forall$ *h*, In *h* *quorum* $\rightarrow$
 In (*eTerm* *e*, *e*) (*allEntries* (*fst* (*nwState* *net* *h*))).

Definition *committed* (*net* : *network*) (*e* : *entry*) (*t* : *term*) :=

$\exists$ *h* *e'*,
 $eTerm$ *e'* $\leq$ *t* $\wedge$
 $directly_committed$ *net* *e'* $\wedge$
 $eIndex$ *e* $\leq$ $eIndex$ *e'* $\wedge$
 In *e* (log (*snd* (*nwState* *net* *h*))) $\wedge$
 In *e'* (log (*snd* (*nwState* *net* *h*))).

Definition *leader_completeness_directly_committed* *net* :=

$\forall$ *t* *e* log *h*,
 $directly_committed$ *net* *e* $\rightarrow$
 $t > eTerm$ *e* $\rightarrow$
 In (*t*, log) (*leaderLogs* (*fst* (*nwState* *net* *h*))) $\rightarrow$
 In *e* log .

Definition *leader_completeness_committed* *net* :=

$\forall$ *t* *t'* *e* log *h*,
 $committed$ *net* *e* *t* $\rightarrow$
 $t' > t \rightarrow$
 In (*t'*, log) (*leaderLogs* (*fst* (*nwState* *net* *h*))) $\rightarrow$
 In *e* log .

Definition *leader_completeness* *net* :=

$leader_completeness_directly_committed$ *net* $\wedge$
 $leader_completeness_committed$ *net*.

VARIABLE *committed*

In *AdvanceCommitIndex*(*i*), for each newly committed *idx*:
 $committed' = committed \cup$
 $\{[index \mapsto idx,$
 $entry \mapsto log[i][idx],$
 $cterm \mapsto currentTerm[i],$
 $cleader \mapsto i,$
 $cquorum \mapsto Agree(idx)]$
 $: idx \in (commitIndex[i] + 1) .. newCommitIndex\}$

VARIABLE *elections*

In *BecomeLeader*(*i*), when candidate *i* wins its term:
 $elections' = elections \cup$
 $\{[eterm \mapsto currentTerm[i],$
 $eleader \mapsto i,$
 $elog \mapsto log[i],$
 $evotes \mapsto votesGranted[i],$
 $evoterLog \mapsto voterLog[i]]\}$

LeaderCompleteness $\triangleq$

$\forall c \in committed :$
 $\forall e \in elections :$
 $c.cterm < e.eterm \Rightarrow$
 $\wedge c.index \in 1 .. Len(e.elog)$
 $\wedge e.elog[c.index] = c.entry$

Figure: Leader Completeness in Verdi Raft (Coq) and abstract Raft (TLA⁺), pretty-printed. In (a), *e'* is the commit certificate and its term *eTerm* *e'* is the commit term, which *t* bounds from above. In (b), the history variables *committed* and *elections* both start empty, and *c.cterm* is the commit term. Both commit to the *commit-term* reading of Ongaro & Ousterhout's Figure 3, and in both the comparison with the later leader's term is strict.